

10. The Equator, Tropic of Cancer and Tropic of Capricorn, all the three pass through which one of the following Continents ?

- (a) Africa (b) Asia
(c) North America (d) South America

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (a)

The Equator, Tropic of Cancer and Tropic of Capricorn all the three pass through the African continent. The Equator and the Tropic of Capricorn pass through the South American continent.

11. Intertropical Convergence Zone (ITCZ), a low-pressure zone, is located.

- (a) between trade winds and westerlies belt.
(b) between westerlies and polar winds belt
(c) nearby the poles
(d) at the equator

M.P. P.C.S. (Pre) 2018

Ans. (d)

The Intertropical Convergence Zone is a low-pressure zone located at the equator.

iv. The Tropic of Cancer

*The Tropic of Cancer is **23°30'** degrees latitude north of the equator. Due to the axial tilt of the earth and its revolution, the Tropic of Cancer (**23°30'N** latitude) starts getting oriented towards the sun by 21st March. *Three months later i.e. by **21st June** the Tropic of Cancer is aligned in a way that it receives the vertical rays of the Sun. Hence **21st June** is known as the day of the summer solstice in Northern Hemisphere. *The Tropic of Cancer passes through the following countries/regions of the world—

- (1) Algeria
- (2) Niger
- (3) Libya
- (4) Egypt
- (5) Saudi Arabia
- (6) United Arab Emirates
- (7) Oman
- (8) India
- (9) Bangladesh
- (10) Myanmar
- (11) China
- (12) Taiwan
- (13) Mexico
- (14) Bahamas
- (15) Mauritania
- (16) Mali
- (17) Western Sahara

22nd December, when the sun is vertically overhead of The Tropic of Capricorn it is called winter solstice. Since the Earth is inclined at an angle of 23°30', when the sun's rays fall at a perpendicular angle (90°) at the Tropic of Capricorn, then at the same time the Tropic of cancer will witness an inclination of 90°–47°= 43°. *The angular difference between the Tropic of Cancer and the Tropic of Capricorn is 23°30' + 23°30' = 47°.

1. At the Winter Solstice the noon altitude of the Sun at the Tropic of Cancer is :

- (a) 23.5° (b) 43.0°
(c) 47.0° (d) 66.5°

U.P.P.C.S. (Pre) 1997

Ans. (b)

The winter solstice falls around 22 December when rays of the Sun fall perpendicular upon the Tropic of Capricorn because the Earth is tilted at its axis by 23°30'. Therefore when at the Tropic of Capricorn the elevation of the Sun is 90°, the elevation of the Sun at the Tropic of Cancer will be 90° – 47° = 43° at the same time. The difference between the Tropic of Cancer and Tropic of Capricorn is 23°30' + 23°30' = 47°.

2. The Tropic of Cancer does not pass through –

- (a) Egypt (b) India
(c) Iran (d) Myanmar

U.P.P.C.S. (Pre) 2005

Ans. (c)

The Tropic of Cancer passes through the following countries/regions of the world –

Mexico	Saudi Arabia
Bahamas	United Arab Emirates
Mauritania	Oman
Mali	India
Algeria	Bangladesh
Niger	Myanmar
	China
Libya	Taiwan
Egypt	Western Sahara

Hence, it is clear that the Tropic of Cancer does not pass through Iran. Therefore (c) is the correct answer.

v. The Tropic of Capricorn

The 23°30' South latitude in the Southern Hemisphere is known as the Tropic of Capricorn. On 21/22 December the sun shines vertically overhead the Tropic of Capricorn, and it is the longest day in the Southern Hemisphere receiving sunlight for the maximum duration. *During this time there is low air pressure in the Southern Hemisphere and thus the

region experiences summer. On in contrast the Northern Hemisphere has high air pressure and experiences winter. During this time the sun's rays fall at an oblique angle on the Tropic of Cancer. *The Tropic of Capricorn passes through the following countries/regions:

- (1) Namibia
- (2) Australia
- (3) Chile
- (4) Argentina
- (5) Paraguay
- (6) Brazil
- (7) Tonga
- (8) French Polynesia (France)
- (9) Botswana
- (10) Mozambique
- (11) South Africa
- (12) Madagascar.

1. Which of the following statements is true regarding earth movement?

1. Earth rotates on its axis from west to east resulting into seasonal changes.
2. Earth revolution around sun results into occurrence of day and night on earth.

- (a) Only 1 is true (b) Both 1 and 2 are true
(c) Only 2 is true (d) Both 1 and 2 are false

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

Earth has seasons because its axis of rotation is tilted at an angle of 23.5° .

Hence statement 1 is incorrect.

It is worth mentioning that night and day occur due to the Earth's rotation on its axis.

Hence statement 2 is also incorrect.

2. Tropic of Capricorn does not pass through –

- (a) Bolivia (b) Paraguay
(c) Argentina (d) Brazil

U.P. Lower Sub. (Pre) 1997

Ans. (a)

Tropic of Capricorn passes through the following countries/regions of the world-

Brazil	Australia
Paraguay	Madagascar
Argentina	Mozambique
Chile	South Africa
French Polynesia (France)	Tonga
Botswana	Namibia

From the above list, it is clear that Tropic of Capricorn does not pass through Bolivia.

3. Think about the following countries –

1. Australia 2. Namibia
3. Brazil 4. Chile

Tropic of Capricorn passes through which of the following countries.

Code :

- (a) Only 1 (b) 2, 3, and 4
(c) 1, 2 and 3 (d) 1, 2, 3 and 4

U.P.P.C.S. (Pre) 2003

Ans. (d)

See the explanation of the above question.

4. Which of the following phenomenon happens in India when the sun-shines vertically over the Tropic of Capricorn?

- (a) High pressure develops over North-West India
(b) Low pressure develops over North-West India
(c) No change in pressure occurs over North-West India
(d) Temperature increases in North-West India
(e) None of these

Chhattisgarh P.C.S. (Pre) 2015

Ans. (a)

When the sun shines vertically over the Tropic of Capricorn, the northwestern part of India becomes colder than the Arabian Sea and high pressure develops over northwest India.

vi. Day & Night

*The earth has two kinds of movements – (1) Rotation i.e. rotation of Earth around its own axis. (2) Revolution i.e. the movement of the earth around the sun. *The earth on its axis rotates from west to east. The rotation of the earth causes the phenomena of Day and Night. The earth is tilted on its axis at an angle of 23.5° as a result of which the duration of Day and Night is not always equal. *However, the duration of Day and Night is always equal on the equator as the sun always shines vertically overhead on the equator. *Between **21st March** and **23rd September** the sun shines between the equator and the Tropic of Cancer, as a result of which the Northern Hemisphere receives more than 12 hours of sunlight. Thus the Days are longer and nights are shorter in the Northern Hemisphere during this time.

*Similarly, from **23rd September** to **21st March**, the sun shines between the Equator and the Tropic of Capricorn. As a result the Southern Hemisphere receives more than 12 hours of sunlight. As one moves closer to the South Pole the duration of Day keeps on increasing, during this period the South Pole witnesses a 6 month long daytime.

*Earth revolves around the sun in an elliptical orbit and completes one revolution around the sun in approx 365 days and 6 hours. Thus because of revolution and axial tilt, the angle of insolation changes throughout the year. Solstices occur twice a year.

***Summer solstice** – On 20/21st June the sun shines vertically overhead at Tropic of Cancer. This phenomenon is called the summer solstice. On 20/21st June the Northern Hemisphere experiences the lengthiest day. The phenomenon is also called **June solstice**. During this time the Southern Hemisphere witnesses the shortest day.

***Winter solstice** – On 21/22nd December, the sun shines vertically overhead at the Tropic of Capricorn. This phenomenon is called the winter solstice. During this time the Southern Hemisphere witnesses the lengthiest day and Northern Hemisphere witnesses the shortest day. This phenomenon is also known as the **December solstice**.

***On 20/21st March The Vernal Equinox** occurs and on **22/23rd September The Autumnal Equinox**. During this time the length of day and night is equal. Due to the revolution and axial tilt of the Earth, the Tropic of Cancer (23.5° North Latitude) in the Northern Hemisphere starts getting oriented towards the sun from 21 March and by 21st June the sun is vertically overhead the Tropic of Cancer.

*The size of the shadow of any object depends upon the size of the object, the distance from the sun and the angle it creates with respect to the rays of the sun.

*The size of the shadow will be the smallest when the angle is least.

*When the angle created is 0° there will be no shadow formation. *The shadow formation in the Northern Hemisphere will be at a minimum on 21st June.

1. On which date day and night are of equal duration?

- (a) 22 June (b) 23 September
- (c) 22 December (d) 21 February

M.P.P.C.S. (Pre) 1992

Ans. (b)

Day and night are of equal duration on the day of the equinox. Equinox literally means "equal day and equal night". An equinox is an astronomical event in which Sun is vertically overhead of the equator, which occurs twice each year, i.e., 20/21st March and 22/23rd September. Therefore, in the given options 23rd September is the correct answer.

2. Equinox, the two periods in the year when day and night are equal, occurs on :

- (a) 21 March and 23 September
- (b) 22 February and 23 August
- (c) 15 October and 23 April
- (d) 22 July and 22 December

U.P. Lower Sub. (Spl) (Pre) 2004

U.P.P.C.S. (Pre) 1994

Ans. (a)

See the explanation of the above question.

3. Where are day and night equal throughout the year?

- (a) North Pole (b) South Pole
- (c) Equator (d) Nowhere

M.P.P.C.S. (Pre) 1995

Ans. (c)

Day and night are of equal duration on The Equator (12 hours each). The duration of day and night varies in the direction of north or south to the equator. There is no winter season on the equator. It only experiences wet and dry weather.

4. At the time the summer solstice in the northern hemisphere the length of the day will be 12 hours at the :

- (a) Tropic of Cancer (b) Tropic of Capricorn
- (c) Arctic Circle (d) Equator

U.P.P.C.S. (Pre) 1997

Ans. (d)

Two types of solstices occur in a year (i) Summer Solstice and (ii) Winter Solstice. The sun reaches the Tropic of Cancer in the month of June in the Northern Hemisphere and the Summer Solstice takes place, just opposite to it in the Southern Hemisphere the Winter Solstice takes place. At the time of Summer Solstice the longest day and shortest night occur in Northern Hemisphere, but in the Southern Hemisphere the shortest day and longest night occur. At the same time, there is not much difference in days and nights at the Equator. Therefore option (d) is the correct answer.

5. In the northern hemisphere, the longest day of the year normally occurs in the :

- (a) First half of the month of June
- (b) Second half of the month of June
- (c) First half of the month of July
- (d) Second half of the month of July

I.A.S. (Pre) 2022

Ans. (b)

The longest day of the year in the Northern Hemisphere usually occurs in the second fortnight of June. This usually occurs on the date of 20 or 21 June. And in the Southern Hemisphere, the longest day of the year occurs in the second fortnight of December.

6. The shortest day of the year in Northern hemisphere is on –

- (a) 21 December (b) 22 December
(c) 21 June (d) 22 June

47th B.P.S.C. (Pre) 2005

M.P.P.C.S. (Pre) 1990

Ans. (b)

The shortest day of the year in Northern Hemisphere occurs during the winter solstice on 21/22 December. Contrary to it on 20/21 June longest day occurs in the Northern Hemisphere during the summer solstice. Opposite to it, in the Southern Hemisphere, the shortest day occurs on 20/21 June and longest day on 21/22 December.

7. Which is the longest day of the year?

- (a) 21 May (b) 21 June
(c) 22 December (d) 25 December

M.P.P.C.S. (Pre) 2015

Ans. (*)

See the explanation of the above question.

8. Which of the following is the longest day of the year ?

- (a) 22 July (b) 25 December
(c) 21 June (d) 23 March

U.P.P.C.S. (Pre) 1992

Ans. (c)

See the explanation of the above question.

9. The Longest day in southern Hemisphere is

- (a) 22 June (b) 22 December
(c) 21 March (d) 22 September

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

See the explanation of the above question.

10. On which of the following dates your shadow at noon is the smallest ?

- (a) December 25 (b) March 21
(c) June 21 (d) February 14

U.P.P.C.S. (Pre) 2006

Ans. (c)

In the course of a year, the Earth's revolution around the Sun with its axis inclined at $66\frac{1}{2}^\circ$ to the plane of the ecliptic changes the apparent altitude of the midday Sun. After the March equinox, the sun appears to move north and is vertically overhead at the Tropic of Cancer ($23\frac{1}{2}^\circ$) on about 21 June (summer solstice). Since the Sun is vertically overhead at noon on 21st June, 0th shadow is formed.

11. Summer Solstice occurs every year on –

- (a) 23 September (b) 21 March
(c) 4 July (d) 21 June

U.P.P.C.S. (Pre) 2010

Ans. (d)

See the explanation of the above question.

12. On 21st June, the Sun :

- (a) does not set below the horizon at the Arctic Circle
(b) does not set below the horizon at Antarctic Circle
(c) shines vertically overhead at noon on the Equator
(d) shines vertically overhead at the Tropic of Capricorn

I.A.S. (Pre) 2019

Ans. (a)

The Sun is directly overhead at 'high noon' on the equator twice per year, at the equinoxes. On the Arctic Circle, the Sun does not set at all on the summer solstice which occurs on 21st June. On that one day, the Sun traces a complete circle just above the horizon as the Earth rotates.

13. On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours?

- 1. Equator 2. Tropic of Cancer**
3. Tropic of Capricorn 4. Arctic Circle

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) 3 and 4 (d) 2 and 4

I.A.S. (Pre) 2024

Ans. (d)

On June 21 every year, among the given latitudes, the Tropic of Cancer and the Arctic Circle (as these are in Northern Hemisphere) experience a sunlight of more than 12 hours. On 21st June, the Northern Hemisphere is tilted towards the Sun. The rays of the Sun fall directly on the Tropic of Cancer. As a result, these areas receive more heat. The areas near the poles receive less heat as the rays of the Sun are slanting. The North Pole is inclined towards the Sun and the places beyond the Arctic Circle experience continuous daylight for about six months. Since a large portion of the Northern Hemisphere is getting more light from the Sun, it is summer in the regions north of the equator. The longest day and the shortest night at these places occur on around 21st June. This position of the Earth in Northern Hemisphere is called the Summer Solstice. At this time in the Southern Hemisphere all these conditions are reversed. It is winter season there. The nights are longer than the days.